D SERIES

PROFESSIONAL CLASS-D POWER AMPLIFIER

Models: D200 · D300 · D4200 · D4300

USER MANUAL

D200	D300	D4200	D4300

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1. Introduction & Overview

Thank you for choosing the Louis Martin Audio D Series Professional Power Amplifier. The D Series represents the pinnacle of Class-D amplification technology, meticulously engineered to deliver exceptional reliability and sonic performance in demanding professional audio environments. Whether you are powering a live sound reinforcement system, a fixed installation, or a touring rig, the D Series provides the headroom, efficiency, and control you need.

The D Series lineup includes four models — D200, D300, D4200 and D4300 — covering both 2-channel and 4-channel configurations with power outputs from 200 W to 600 W per channel at 4Ω. All models share the same high-efficiency switching power supply architecture, advanced DSP limiting, and automatic thermal management.

Key Features

Feature	Description
Class-D Topology	Ultra-high efficiency, minimal heat generation and power loss
Auto Temperature Control	Variable fan speed adjusts automatically with load and ambient temperature
Advanced Limiting	Multi-stage clip, thermal and DC protection for amp and loudspeakers
Balanced XLR I/O	20 kΩ balanced inputs reject common-mode noise and hum
Bridgeable Channels	2-ch models bridge to mono; 4-ch models bridge in pairs
THD+N < 0.05%	Exceptional low distortion at 1 kHz, 8Ω
S/N Ratio > 98 dB	A-weighted; ultra-quiet background for audiophile clarity
Damping Factor > 500	Tight loudspeaker control at 1 kHz / 8Ω

2. Safety Instructions

Please read all safety instructions carefully before operating this unit. Retain this manual for future reference.

■ **ELECTRIC SHOCK HAZARD**	Do not open the unit. No user-serviceable parts inside. Refer all servicing to qualified technicians.
■ **WATER & MOISTURE**	Do not use this product near water, rain, or in humid environments. Keep liquids away at all times.
■ **VENTILATION**	Do not block any ventilation openings. Maintain at least 1U (44 mm) of free space above and below the unit in a rack.
■ **HEAT SOURCES**	Do not install near radiators, heat registers, stoves, or other heat-generating apparatus.
■ **POWER CORD**	Protect the power cord from being walked on or pinched. Do not use damaged cords.
■ **LIGHTNING**	Unplug the unit during lightning storms or when unused for long periods.
■ **CLEANING**	Clean only with a dry cloth. Never use solvents, sprays, or abrasive cleaners.
■ **FUSING**	Replace fuses only with the specified type and rating. Never bypass or bridge fuses.
■ **SERVICING**	Servicing is required when the unit is damaged mechanically or electrically, or after use in a smoky or dusty environment.
■ **VOLUME LEVELS**	Exposure to high sound pressure levels can cause permanent hearing damage. Use appropriate gain structure.

3. Package Contents

Please verify all items are present upon unpacking. Contact your dealer immediately if any item is missing or damaged.

✓	D Series Power Amplifier (as ordered model)	1 pc
✓	IEC AC Power Cord (country-specific)	1 pc
✓	Rack-mount ear brackets (pre-installed)	1 set
✓	M6 rack screws and cage nuts	4 pcs
✓	Quick-Start Guide	1 pc
✓	User Manual (this document)	1 pc

4. Panel Layout & Controls

4.1 Front Panel

- **Power Switch:** Main power ON/OFF rocker switch with illuminated indicator.
- **Power LED (Blue):** Illuminates blue when the unit is powered on and ready.
- **Signal LED (Green):** Illuminates green when an audio signal is detected on the channel.
- **Clip LED (Red):** Illuminates red when the channel approaches clipping. Reduce input gain immediately.
- **Protect LED (Orange):** Illuminates when a protection circuit is active (thermal, DC, or short-circuit).
- **Volume Knobs:** Individual channel gain controls (0 dB = fully clockwise). Rotate counter-clockwise to reduce gain.

4.2 Rear Panel

- **XLR Input Connectors:** Balanced XLR-F inputs, one per channel. Pin 1 = Ground, Pin 2 = Hot (+), Pin 3 = Cold (–).
- **Link Output (XLR-M):** Buffered signal pass-through to daisy-chain additional amplifiers.
- **Binding Post / Speakon Outputs:** Speaker outputs per channel. Colour-coded: Red = +, Black = –.
- **Bridge / Stereo Switch:** Toggles channel pair between Stereo and Bridged Mono mode (D200/D300). For D4200/D4300, pairs CH1+CH2 and CH3+CH4.
- **IEC AC Inlet:** Standard IEC C13/C14 inlet. 220–240 V AC, 50/60 Hz.
- **Fuse Holder:** Accessible fuse holder — see rear panel label for correct fuse rating. Replace with same type only.

5. Technical Specifications

Parameter	D200	D300	D4200	D4300
Power @ 8Ω / ch	250 W × 2	350 W × 2	200 W × 4	300 W × 4
Power @ 4Ω / ch	400 W × 2	600 W × 2	350 W × 4	500 W × 4
Bridge @ 8Ω	700 W	1000 W	650 W × 2	900 W × 2
Avg. Current Draw	2.5 A @ 230 V	3 A @ 230 V	4.5 A @ 230 V	5 A @ 230 V
Max Power Consumption	575 W	690 W	1035 W	1150 W
Frequency Response	20 Hz – 20 kHz (±0.5 dB)	←	←	←
THD+N	< 0.05% @ 8Ω, 1 kHz	←	←	←
S/N Ratio	> 98 dB (A-weighted)	←	←	←
Damping Factor	> 500 (1 kHz @ 8Ω)	←	←	←
Crosstalk	> 80 dB @ 1 kHz	←	←	←
Input Sensitivity	0.775 V / 1.44 V	←	←	←
Input Impedance	20 kΩ Balanced	←	←	←
Dimensions (W×D×H)	482×220×48 mm	←	482×330×88 mm	←
Net Weight	2.5 kg	2.5 kg	4.8 kg	5.0 kg

■ *← indicates the value is identical to the D200 column for that parameter.*

6. Installation & Rack Mounting

6.1 Rack Mounting

The D Series amplifiers are designed for standard 19-inch rack enclosures. The D200 and D300 occupy 1U (48 mm height), while the D4200 and D4300 occupy 1U (48 mm height). Always use the supplied M6 rack screws and cage nuts for secure mounting.

- 1. Install cage nuts in the rack at the correct U positions for your model.
- 2. With a second person assisting, carefully slide the amplifier into the rack rails.
- 3. Align the front panel ear holes with the cage nuts and hand-tighten the M6 screws.
- 4. Tighten all four screws firmly with a screwdriver — do not over-torque.
- 5. Ensure at least 1U (44 mm) of free ventilation space above the amplifier. Never stack amplifiers without a ventilation gap.
- 6. For touring use, consider rear rack support rails to bear the chassis weight and prevent connector stress.

6.2 Ventilation Requirements

The D Series uses a variable-speed rear-to-front (or front-to-rear depending on variant) fan cooling system. For optimal performance and longevity, ensure the following:

- Minimum 1U (44 mm) clear space directly above the unit.
- Ambient operating temperature: 0°C to 40°C. Do not exceed 40°C rack ambient.
- Do not block the rear exhaust or front intake vents with cables or panels.
- In high-dust environments, inspect and clean the fan filters every 3 months.

6.3 Power Requirements

All D Series models operate on 220–240 V AC, 50/60 Hz. Ensure the mains circuit supplying the amplifier is adequately rated:

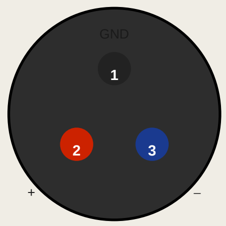
Model	Max Draw (A)	Min Circuit (A)	Recommended MCB
D200	2.5 A	10 A	10 A Type B
D300	3.0 A	10 A	10 A Type B
D4200	4.5 A	10 A	13 A Type B
D4300	5.0 A	10 A	13 A Type B

7. Connection Wiring Guide

This section covers all standard connection scenarios for the D Series amplifier. Always power down the amplifier before making or changing any connections. Set all gain controls to minimum (fully counter-clockwise) before powering up.

7.1 XLR Input Pinout

All D Series models use standard balanced XLR-F (female) inputs. The pin assignment follows the international AES/EBU standard:



XLR BALANCED INPUT

Pin	Function	Wire Color (typical)
1	Ground / Shield	Black or bare
2	Signal Hot (+)	Red
3	Signal Cold (–)	White or Blue

7.2 Stereo 2-Channel Wiring (D200 / D300)

In stereo mode, each channel operates independently. Connect the left source output to CH1 input and the right source output to CH2 input. Connect the corresponding speaker to each channel's output binding posts.

STEREO MODE — CONNECTION DIAGRAM

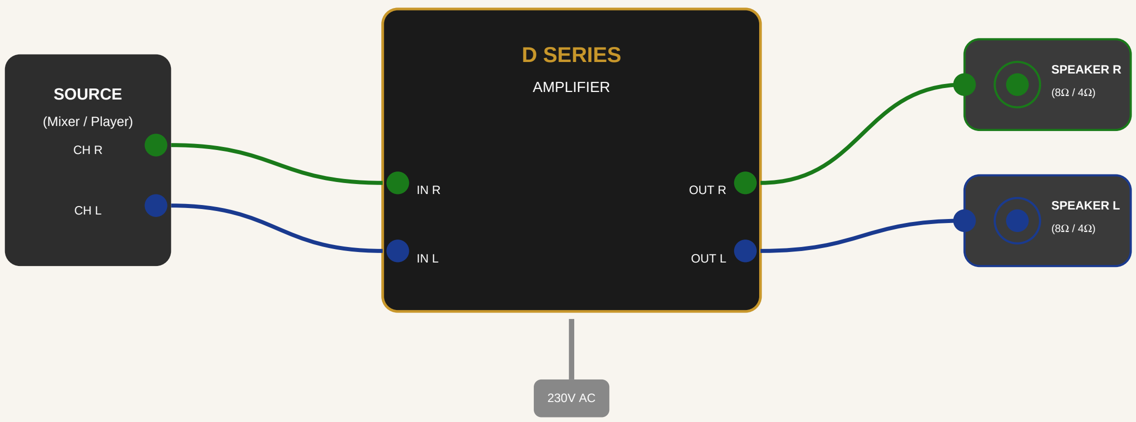


Figure 1 — Stereo 2-channel connection (D200/D300)

- 1. Set both volume knobs to minimum (fully counter-clockwise).
- 2. Connect the balanced XLR output from your mixer/source to CH1 INPUT (left) using a quality XLR cable.
- 3. Connect the balanced XLR output to CH2 INPUT (right).
- 4. Connect the CH1 output binding post (+) to the speaker positive terminal using suitable speaker cable (minimum 1.5 mm²).

- 5. Connect the CH1 output binding post (–) to the speaker negative terminal.
- 6. Repeat steps 4–5 for CH2 and the second speaker.
- 7. Power on the amplifier and slowly raise the gain controls to the desired operating level.

7.3 Bridged Mono Wiring (D200 / D300)

Bridged mono mode combines both channels into a single high-power output, approximately doubling the voltage swing. **Use bridged mode with speakers rated at minimum 8Ω only.** Never connect a 4Ω load in bridge mode.

■ IMPORTANT: Bridged mode minimum load impedance is 8Ω. A 4Ω load WILL damage the amplifier.

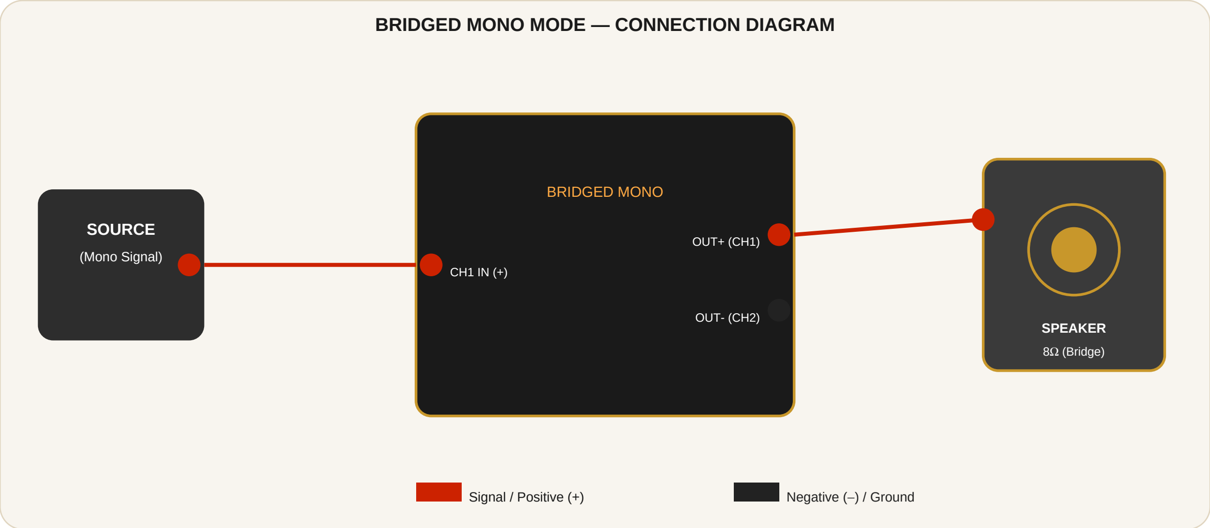


Figure 2 — Bridged mono connection (D200/D300)

- 1. Set the BRIDGE/STEREO switch on the rear panel to BRIDGE position.
- 2. Connect your mono source signal to **CH1 INPUT only**. CH2 input is not used in bridge mode.
- 3. Connect the speaker positive (+) terminal to the CH1 OUTPUT binding post (+) [Red].
- 4. Connect the speaker negative (–) terminal to the CH2 OUTPUT binding post (+) [Red]. Do NOT use the (–) Black terminal.
- 5. Power on and set the CH1 volume to desired level. The CH2 knob has no effect in bridge mode.

7.4 4-Channel Stereo Wiring (D4200 / D4300)

The D4200 and D4300 provide four independent amplifier channels. Each channel has its own XLR input, volume knob, and speaker output. Channels can be used independently or grouped for larger systems.

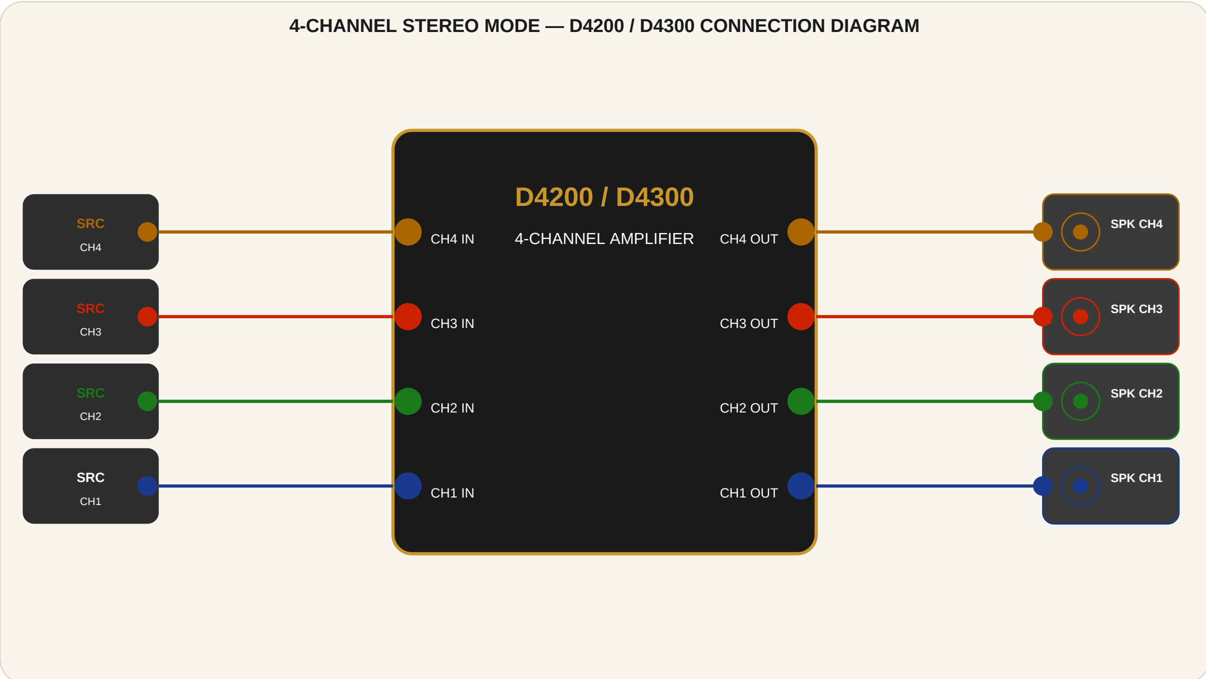


Figure 3 — 4-channel stereo connection (D4200/D4300)

7.5 Daisy-Chaining Multiple Amplifiers

The Link (Thru) XLR-M output on the rear panel provides a buffered copy of the input signal. This allows multiple D Series amplifiers to be driven from a single source without signal degradation.

- Connect the source to the first amplifier's INPUT.
- Connect the LINK OUTPUT of the first amplifier to the INPUT of the second amplifier.
- Continue daisy-chaining up to 8 units from a single source (signal integrity dependent on cable quality).

■ *Use high-quality, shielded XLR cables of 5 m or less for each daisy-chain link to minimize noise.*

7.6 Speaker Cable Guidelines

Run Length	Min. Cable CSA	Recommended CSA	Max. Resistance
Up to 5 m	1.5 mm²	2.5 mm²	< 0.5 Ω
5 m – 15 m	2.5 mm²	4.0 mm²	< 0.5 Ω
15 m – 30 m	4.0 mm²	6.0 mm²	< 0.5 Ω
> 30 m	6.0 mm²		Consult installer

8. Operating Instructions

8.1 Power-On / Power-Off Sequence

Always follow the correct power sequence to protect both the amplifier and connected loudspeakers:

Power-On sequence:

- 1. Sources and signal processors first → Mixers and EQ → D Series Amplifier last.

Power-Off sequence:

- 1. D Series Amplifier first → Mixers and EQ → Sources and signal processors last.

■ *The amplifier incorporates a soft-start relay that mutes the output for approximately 3 seconds after power-on to eliminate switch-on transients.*

8.2 Gain Structure

Proper gain structure is essential for achieving maximum dynamic range with minimum noise. The D Series accepts input sensitivities of 0.775 V (–2.2 dBu) and 1.44 V (+3.2 dBu). Follow this procedure:

- 1. Set the D Series volume knobs to the 12 o'clock (50%) position.
- 2. At the source (mixer), bring up the main output until the clip LEDs on the amplifier just begin to flicker — then back off 3 dB.
- 3. For maximum headroom, use the 1.44 V sensitivity setting and drive the source to +3 dBu nominal level.
- 4. In noisy RF environments, use the lower 0.775 V sensitivity and rely on the mixer's output pad.
- 5. Never operate with the volume knobs at maximum and the source gain at minimum — this degrades the signal-to-noise ratio.

8.3 LED Indicator Reference

LED	Colour	State	Meaning
POWER	Blue	Steady ON	Unit powered and operational
SIGNAL	Green	Flashing	Audio signal present on channel
CLIP	Red	Flashing	Output near clipping — reduce gain
CLIP	Red	Steady ON	Sustained clipping — reduce gain immediately
PROTECT	Orange	Steady ON	Protection active — see Troubleshooting

9. DSP & Limiting Functions

The D Series incorporates a multi-stage protection and limiting system designed to safeguard both the amplifier and connected loudspeakers under all operating conditions.

- **Clip Limiter:** Monitors the output waveform and smoothly reduces gain before hard clipping occurs, preserving transient response while preventing distortion.
- **Thermal Protection:** Continuously monitors the internal heat sink temperature. At 70°C the fan speed increases to maximum. At 85°C the output is muted until the unit cools below 70°C. The PROTECT LED illuminates.
- **DC Output Protection:** Immediately disconnects the speaker outputs if a DC offset is detected at the output, protecting loudspeakers from damage.
- **Short-Circuit Protection:** Detects dead short at the speaker terminals and disables the output. After the fault is cleared, cycle power to restore normal operation.
- **Inrush Current Limiting:** A soft-start thermistor limits the inrush current at power-on, protecting the mains fuse and relay contacts.
- **Over-Voltage Protection:** Shuts down the amplifier if the mains supply exceeds the rated voltage by more than 15%.

10. Troubleshooting

Symptom	Possible Cause	Solution
No power / no LED	Mains fuse blown; power cord disconnected	Check IEC cord; inspect and replace fuse with correct rating
No audio output	Volume at minimum; PROTECT active; input cable fault	Raise volume; check PROTECT LED; reseat XLR cables
PROTECT LED on	Thermal overload; DC fault; short circuit	Power off, allow 10 min cool-down, check speaker wiring for shorts, then restart
Excessive hum / noise	Ground loop; unbalanced cable near mains wiring	Use balanced XLR cables; add ground-lift adapter on console; separate signal from power cables
Distorted output	Clipping (red LED active); gain too high	Reduce source gain or amp volume; check gain structure
Fan very loud	High thermal load; blocked ventilation	Reduce drive level or add ventilation; check for blocked vents
One channel silent	Input cable fault; CH volume at zero	Check input cable; verify volume knob is raised
Unit shuts off	Thermal protection triggered	Allow cooling, increase rack ventilation, reduce ambient temperature

■ *If the issue persists after following the steps above, do not attempt to open or repair the unit yourself. Contact an authorised Louis Martin Audio service centre.*

11. Maintenance & Care

11.1 Routine Maintenance Schedule

Interval	Task
Weekly	Visual inspection of all cable connections and connector security.
Monthly	Wipe exterior with a dry cloth; check all LEDs function correctly during operation.
3 Monthly	Inspect fan intake filter for dust accumulation; clean with compressed air if necessary.
Annually	Full professional service inspection recommended for touring units or high-usage installations.

11.2 Storage

- Store in original packaging when not in use for extended periods.
- Storage environment: –20°C to +60°C, relative humidity < 90% non-condensing.
- Keep away from solvents, corrosive substances, and strong magnetic fields.

12. Warranty & Support

Louis Martin Audio warrants this product against defects in materials and workmanship for a period of **2 years** from the date of original purchase by the end user. This warranty covers manufacturing defects only and does not cover damage caused by misuse, accidents, unauthorised modification, or failure to follow the instructions in this manual.

To obtain warranty service:

- Retain proof of purchase (receipt or invoice).
- Contact your authorised Louis Martin Audio dealer or our support team directly.
- Do not ship the unit without prior authorisation — obtain an RMA number first.

Contact	Details
Website	www.louismartinaudio.it
E-mail	contact@louismartinaudio.it
Support Hours	Monday – Friday, 09:00 – 18:00